step 1:Library installation and setup

```
# Install required libraries
!pip install nltk spacy
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
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Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
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Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.21.1)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
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Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.1
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
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Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (25.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.
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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spac
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Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->sp
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2-
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->s
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->spacy) (3.0.3)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1->weasel<0.5.0,>=0
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 113.4 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

step 2:Import libraries

```
import nltk
import spacy
from nltk.corpus import reuters
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer, SnowballStemmer, WordNetLemmatizer
```

step 3:Download NLTK resources

```
nltk.download('reuters')
nltk.download('punkt')
nltk.download('wordnet')
nltk.download('omw-1.4')
```

```
[nltk_data] Downloading package reuters to /root/nltk_data...
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
```

```
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
True
```

step 4:Load news articles

```
# Load one short Reuters news article
doc_id = reuters.fileids()[0]
news_text = reuters.raw(doc_id)

print("News Article Sample:\n")
print(news_text[:500])
```

```
News Article Sample:

ASIAN EXPORTERS FEAR DAMAGE FROM U.S.-JAPAN RIFT
  Mounting trade friction between the
  U.S. And Japan has raised fears among many of Asia's exporting
  nations that the row could inflict far-reaching economic
  damage, businessmen and officials said.
      They told Reuter correspondents in Asian capitals a U.S.
  Move against Japan might boost protectionist sentiment in the
  U.S. And lead to curbs on American imports of their products.
      But some exporters said that while the conflict wo
```

step 5:Tokenization using NLTK

```
nltk.download('punkt_tab')
nltk_tokens = word_tokenize(news_text)
print("NLTK Tokens (first 30):\n")
print(nltk_tokens[:30])
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
NLTK Tokens (first 30):

['ASIAN', 'EXPORTERS', 'FEAR', 'DAMAGE', 'FROM', 'U.S.-JAPAN', 'RIFT', 'Mounting', 'trade', 'friction', 'between', 'the', 'U.S.'
```

step 6:Tokenization using spacy

```
nlp = spacy.load("en_core_web_sm")
doc = nlp(news_text)

spacy_tokens = [token.text for token in doc]
print("spaCy Tokens (first 30):\n")
print(spacy_tokens[:30])
```

```
spaCy Tokens (first 30):

['ASIAN', 'EXPORTERS', 'FEAR', 'DAMAGE', 'FROM', 'U.S.-JAPAN', 'RIFT', '\n  ', 'Mounting', 'trade', 'friction', 'between', 'the'
```

step 7:Stemming

```
porter = PorterStemmer()
porter_stems = [porter.stem(word) for word in nltk_tokens[:30]]
print("Porter Stemmer Output:\n")
print(porter_stems)
```

```
Porter Stemmer Output:

['asian', 'export', 'fear', 'damag', 'from', 'u.s.-japan', 'rift', 'mount', 'trade', 'friction', 'between', 'the', 'u.s.', 'and'
```

step 7.2:Snowball stemmer

```
snowball = SnowballStemmer("english")
snowball_stems = [snowball.stem(word) for word in nltk_tokens[:30]]
print("Snowball Stemmer Output:\n")
print(snowball_stems)
```

```
Snowball Stemmer Output:

['asian', 'export', 'fear', 'damag', 'from', 'u.s.-japan', 'rift', 'mount', 'trade', 'friction', 'between', 'the', 'u.s.', 'and'
```

step 8:Lemmatization

```
wordnet_lem = WordNetLemmatizer()
wordnet_lemmas = [wordnet_lem.lemmatize(word) for word in nltk_tokens[:30]]
print("WordNet Lemmatization Output:\n")
print(wordnet_lemmas)
```

```
WordNet Lemmatization Output:

['ASIAN', 'EXPORTERS', 'FEAR', 'DAMAGE', 'FROM', 'U.S.-JAPAN', 'RIFT', 'Mounting', 'trade', 'friction', 'between', 'the', 'U.S.'
```

step 8.2:Spacy lemmatization

```
spacy_lemmas = [token.lemma_ for token in doc[:30]]
print("spaCy Lemmatization Output:\n")
print(spacy_lemmas)
```

```
spaCy Lemmatization Output:

['asian', 'exporter', 'fear', 'damage', 'from', 'U.S.-JAPAN', 'RIFT', '\n  ', 'Mounting', 'trade', 'friction', 'between', 'the',
```

Start coding or generate with AI.

Double-click (or enter) to edit